# Supplementary Material for “A Probability-Free Compositional Structure Needs Four Quantum Levels but Five Classical States”

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This Supplementary Material contains details of the proofs, the tables used in the main text, the numerical checks and the hardware protocol. References [S1] to [S3] are listed at the end. Two-qubit Pauli operators are written as two-letter labels, for example XY = $X \otimes Y$ and IZ = I $\otimes$ $Z$, and $X_1$ = XI, $Z_1$ = ZI, $Y_1Z_2$ = YZ.

## S1. One qubit: details

Shadow of the ball. Let $\Pi$ be a plane at distance $\delta$ from the centre and $p$ an affine projection onto $\Pi$ along a unit direction that makes the angle $\theta$ with the normal of $\Pi$. The image of the unit ball is an ellipse in $\Pi$ with semi-axes 1 and $1/|\cos\theta|$, centred at the projection of the centre. The section $\Pi \cap \mathbb{B}$ is a disk of radius $(1 - \delta^2)^{1/2}$. The two sets coincide only if $\delta = 0$ and $\theta = 0$.

Complete positivity. A unital qubit channel acts on Bloch vectors by $r \mapsto \Lambda r$. After rotations on both sides, $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$, and complete positivity holds if and only if $|\lambda_1 \pm \lambda_2| \le |1 \pm \lambda_3|$ [S1, S2]. For $\Lambda = \mathrm{diag}(1, 1, 0)$ the condition fails, and the Choi matrix of this map has the eigenvalue $-1/2$. The orthogonal projection onto the equatorial plane is positive but not completely positive.

One-dimensional fixed sets. Write the idempotent as $e(r) = p_0 + [g\cdot(r - p_0)]\, d$. Then $e(e(r)) = e(r)$ for all $r$ gives $g\cdot d = 1$. The image of $\mathbb{B}$ is the segment $\{p_0 + t\, d\}$, with $t$ ranging over an interval of length $2|g|$, so its length is $2|g||d| \ge 2$, with equality only if $g$ is parallel to $d$. A segment in $\mathbb{B}$ has length at most two, with equality only for diameters. The map is therefore the orthogonal projection onto a diameter, $r \mapsto (n\cdot r)n$.

Lower-dimensional orbits. If the forward orbit of $\rho_0$ spans an affine subspace of dimension two or one, the local monoid acts on $K = \mathrm{aff}\, O(\rho_0) \cap \mathbb{B}$ by affine maps, and two channels agree on the orbit exactly when they agree on $K$. By Lemma 2 of the main text, the fixed set of an idempotent of $K$ is the intersection of $K$ with an affine subspace. On a planar $K$, a non-identity, non-constant idempotent has a line as this subspace; on a one-dimensional $K$, every affine idempotent is the identity or constant. The proof of Theorem 1 uses only these facts, so it covers non-faithful realizations as well.

Commuting dephasings. For unit vectors $n$ and $m$, $D_nD_m(r) = (m\cdot r)(n\cdot m)\, n$ and $D_mD_n(r) = (n\cdot r)(n\cdot m)\, m$. The two agree for all $r$ exactly when $n\cdot m = 0$ or $n = \pm m$. In the first case the product is the constant map $r \mapsto 0$, the channel onto I/2.

The single-qubit Pauli dephasings. In the Pauli basis $D_x$ = diag(1, 1, 0, 0), $D_y$ = diag(1, 0, 1, 0) and $D_z$ = diag(1, 0, 0, 1). They generate the five-element semilattice $\{1, D_x, D_y, D_z, D_xD_y\}$, whose three join-irreducibles are permuted by the single-qubit Clifford group as the full $S_3$. This family satisfies conditions (ii) and (iii) of Definition 1 and fails condition (i) by Lemma 1.

**S2. Two qubits: construction and proof of Theorem 2**

Pauli transfer matrices. For a Pauli operator $P$, $D_P(Q) = Q$ if $Q$ commutes with $P$ and $D_P(Q) = 0$ if $Q$ anticommutes with $P$, for every Pauli operator $Q$. The global depolarization multiplies every non-identity Pauli operator by $q$. All these maps are diagonal in the Pauli basis, so they commute.

Table S1. Non-identity Pauli operators kept by each zero-grade element $D_A$, and their intersection with the Pauli support {IZ, IX, XY, ZY} of $\rho_0$.

| $A$ | **Kept Pauli operators $C(A)$** | **Kept components of $\rho_0$** |
|---|---|---|
| $\varnothing$ | all 15 | IZ, IX, XY, ZY |
| $\{X_1\}$ | IX, IY, IZ, XI, XX, XY, XZ | IZ, IX, XY |
| $\{Z_1\}$ | IX, IY, IZ, ZI, ZX, ZY, ZZ | IZ, IX, ZY |
| $\{Y_1Z_2\}$ | IZ, XX, XY, YI, YZ, ZX, ZY | IZ, XY, ZY |
| $\{X_1, Z_1\}$ | IX, IY, IZ | IZ, IX |
| $\{X_1, Y_1Z_2\}$ | IZ, XX, XY | IZ, XY |
| $\{Z_1, Y_1Z_2\}$ | IZ, ZX, ZY | IZ, ZY |
| $\{X_1, Z_1, Y_1Z_2\}$ | IZ | IZ |

The eight rows of the last column are distinct and non-empty. For an element $\tau^n D_A$, the image of $\rho_0$ is $(\mathrm{I} + \Sigma_Q\, q^n\, c_Q\, Q)/4$, with the sum over the kept components $Q$ of $\rho_0$ and $c_Q$ their coefficients in Eq. (4) of the main text. Two elements with different $A$ have different sets of non-zero components in the image, and two elements with the same $A$ and different $n$ differ in the coefficient $q^n a$ of IZ. Hence the map $(n, A) \mapsto \tau^n D_A$ is injective on $\rho_0$ for every $n$.

Symmetry group. The Clifford unitaries $U_\mathrm{C}$ and $U_\mathrm{T}$ of Eq. (3) are determined up to phase by their action on XI, ZI, IX and IZ. $U_\mathrm{C}$ maps YZ to −XI and $U_\mathrm{T}$ maps YZ to −YZ, and $U_\mathrm{T}$ is the Hadamard gate on the first qubit. Modulo phases they generate a group of order six whose induced permutations of $(x_1, x_2, x_3)$ are all six elements of $S_3$. Twirling over this group sends IX, XY and ZY to (IX + XY + ZY)/3, sends IY, XX and ZX to ±(IY − XX − ZX)/3, fixes IZ and annihilates the remaining non-identity Pauli operators. The $H$-invariant operators are

therefore spanned by I, IZ, $K_1$ and $K_2$. The state $\rho_0 = (\mathrm{I} + a\ \mathrm{IZ} + b\ K_1)/4$ is positive definite when $|a| + 3|b| < 1$, since each Pauli operator has operator norm one; for $a = b = 0.2$ its smallest eigenvalue is 0.15. No atom fixes $\rho_0$.

Normalization and grading. Every element of $H$ commutes with $\tau$, because depolarization is unitarily covariant, and permutes the atoms, so it normalizes the monoid and preserves both the grading and the zero-grade sector. The grading channel must have eigenvalues of modulus strictly between 0 and 1 on all of IZ, IX, XY and ZY. A Pauli channel with eigenvalue $q$ on IZ and 0 on IX, XY and ZY, for example, gives $\tau D_{\varnothing}(\rho_0) = \tau D_{X1}(\rho_0)$ and destroys faithfulness.

### S3. Proposition 1: enumeration and a product symmetry group

Write a two-qubit Pauli operator, up to phase, as a vector in the binary symplectic space $\mathbb{F}_2^4$, with commutation given by the symplectic form. Of the 455 triples of non-identity Pauli operators, 420 are linearly independent. Among them, 60 have a uniform commutation pattern, and all 60 are pairwise anticommuting, since an isotropic subspace has dimension at most two. None of the 60 consists only of operators of the form $P \otimes \mathrm{I}$ or $\mathrm{I} \otimes P$. The enumeration is part of theory_checks.py.

For the triple XI, ZI, YZ, all products of single-qubit Clifford unitaries, with or without the swap, that map the triple to itself induce either the identity or the transposition of XI and ZI. A full $S_3$ therefore requires an entangling element, such as $U_{\mathrm{C}}$. For the triple XX, XY, XZ the situation differs. The script product_observer.py finds a group of six product Clifford unitaries $V \otimes W$ that induces all six permutations of the three atoms, determines its invariant operators, and constructs an invariant state with Pauli support {XI, YX, YY, YZ, ZX, ZY, ZZ} and smallest eigenvalue 0.16. The component XI is kept by every $D_A$, and the intersections of the support with the eight commutants are distinct. With global depolarization as the grading, the 120 elements with $n < 15$ give pairwise distinct images, and the criterion of S2 extends this to all $n$.

The monoid $\mathbb{N}_0 \times M_3$. Among the dependent triples with a uniform commutation pattern, the single-qubit triple XI, YI, ZI generates $M_3$: each pairwise product equals $D_{\mathrm{XI}}D_{\mathrm{YI}}D_{\mathrm{ZI}}$, which keeps exactly the Pauli operators IX, IY and IZ and is therefore not constant on two qubits. With $V_{\mathrm{c}} = \mathrm{SH}$, which maps $X \to Z \to Y \to X$, and $V_{\mathrm{t}} = (X - Y)/\sqrt{2}$, which exchanges $X$ and $Y$ up to sign and maps $Z$ to $-Z$, the product unitaries $V_{\mathrm{c}} \otimes \mathrm{I}$ and $V_{\mathrm{t}} \otimes X$ generate a group that induces all six permutations of the three atoms and fixes $\rho_M = (\mathrm{I} + 0.2\ \mathrm{IX} + 0.2\ (\mathrm{XZ} + \mathrm{YZ} + \mathrm{ZZ}))/4$, whose smallest eigenvalue is 0.15. The support of $\rho_M$ meets the commutants of the five elements of $M_3$ in distinct non-empty sets, {IX, XZ, YZ, ZZ}, {IX, XZ}, {IX, YZ}, {IX, ZZ} and {IX}, and IX is kept by all of them, so with global depolarization the monoid acts

injectively on the forward orbit of $\rho_M$. theory_checks.py verifies the group action, the invariance and 75 distinct images for $n < 15$. Of the 35 dependent triples with a uniform commutation pattern, 15 are commuting and 20 anticommuting; all generate five-element semilattices.

Classical five-state realization (Theorem 3). The parameters satisfy $0 < s < 1$, $0 < u \le 1$ and $s \ne u$, and the distribution has $p > 0$ on each of the states 1, 2, 3 and a weight on state 4 different from $(1 - s)/(1 - s + u)$. The weight $w_n$ on state 4 obeys $w_{n+1} = (1 - s) + (s - u)\, w_n$ for every $A$, so $w_n - w_* = (s - u)^n (w_0 - w_*)$, which separates all $n$. With column-stochastic matrices on the states $\{0, 1, 2, 3, 4\}$, the atom $x_i$ replaces the column of state $i$ by the unit vector of state 0, and $\tau$ has columns $s\, e_k + (1 - s)\, e_4$ for $k = 0, \ldots, 3$ and $u\, e_0 + (1 - u)\, e_4$ for state 4. The script classical_b3.py checks, for $s = 0.6$, $u = 0.3$ and the distribution (0.1, 0.2, 0.2, 0.2, 0.3), idempotence and commutation, the union law, the eight distinct zero-grade elements, the $S_3$ action of the permutations of $\{1, 2, 3\}$, which commute with $\tau$ and fix the distribution, 120 distinct images for $n < 15$, and that the orbit spans the four-dimensional simplex.

Qutrit fixed sets. For a unit vector $v$, the block set of $v$ is the set of states commuting with $|v\rangle\langle v|$, of dimension four. For two block sets with $v \perp w$, the common states commute with the projectors onto $v$, $w$ and the third basis vector and are the diagonal states of that basis, of dimension two. If $v$ and $w$ are neither equal nor orthogonal, the two projectors generate the full matrix algebra on $\mathrm{span}\{v, w\}$, so the common states are proportional to the identity on that span plus a weight on its complement, a segment. A face, the states supported on a two-dimensional subspace $S$, meets the block set of $v$ in the diagonal states of $S$ with respect to $v$ when $v \in S$, a segment, is contained in it when $S = v^{\perp}$, and meets it in at most a segment otherwise. These facts, used in the proof of Theorem 3(ii), are checked numerically in qutrit_pinch.py together with the commutation criterion for pinchings.

Hybrids with four perfectly distinguishable states (Theorem 3(iv)). Throughout, $t \ge 1$ is the dimension of the fixed set of the product of the three atoms in $K$, the atoms need fixed sets of dimension at least $t + 2 \ge 3$, and the pairwise products fixed sets of dimension at least $t + 1 \ge 2$ that differ from each other. Let $c \in H$ induce the 3-cycle $x_1 \to x_2 \to x_3 \to x_1$ on $K$ and let $E_1$ be the ergodic projection of $x_1$, which coincides with $x_1$ on $K$. Then $E_2 = cE_1c^{-1}$ and $E_3 = c^2E_1c^{-2}$ coincide with $x_2$ and $x_3$ on $K$, and their fixed sets $G_i$ lie in one orbit of the automorphism group, with $\mathrm{Fix}_K(x_i) = G_i \cap K$. All conditions below are applied to the $G_i$, whose pairwise and triple intersections contain the corresponding fixed sets in $K$. Some forms carry a binary label that automorphisms preserve or flip, such as which of two blocks is kept whole. If $c$ preserves the label, the three $G_i$ carry the same label. If $c$ flips it, $E_1' = c^3E_1c^{-3}$

also coincides with $x_1$ on $K$ but carries the opposite label, so $\mathrm{Fix}_K(x_1)$ lies in $\mathrm{Fix}(E_1) \cap \mathrm{Fix}(E_1')$; this case is treated separately below.

$M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ (blocks $Q$, $a$, $b$; states of dimension five). The forms of dimension at least three are the qubit block with both outcomes (whole set), the qubit block with one outcome or with a fixed mixture $\omega$ of $a$ and $b$ (dimension four), the qubit block alone, and the diagonal states $\mathrm{diag}_n(\mathrm{Q}) \oplus a \oplus b$ for an axis $n$ (dimension three). Only the last forms have orbits with three elements. Two of them with different axes meet in $\{p\ \mathrm{I}/2 \oplus p_a \oplus p_b\}$, the same set for every pair, so the atoms generate $M_3$.

$M_2 \oplus M_2$ (blocks $Q_0$, $Q_1$; dimension seven). The forms of dimension at least three are: (a) one block whole and the other diagonal in a basis, dimension five; (b) one block whole and the other supported on a line, dimension four; (c) one block whole and the other in a fixed state, dimension four; (d) one block alone, dimension three, with an orbit of two elements; (e) the tied sets $\{p\ \sigma \oplus (1 - p)\ V\sigma V^\dagger\}$ with fixed $p$, dimension three; (f) the diagonal states in a basis of each block, dimension three. For (a), (b) and (c), if $c$ preserves the label the three forms keep the same block whole; then in (a) the pairwise intersections all equal $\{Q_0\ \text{whole} \oplus p_1\ \mathrm{I}/2\}$, in (b) and (c) they all equal the states of $Q_0$ alone, and the atoms generate at most $M_3$. If $c$ flips the label, $\mathrm{Fix}_K(x_1)$ lies in the intersection of a form with $Q_0$ whole and one with $Q_1$ whole: for (b) and (c) this intersection has dimension at most one, which contradicts the required dimension three, and for (a) it is a set of type (f), diagonal in a basis of each block, so the fixed sets in $K$ are of type (f) and the argument for (f) applies to them. In (e) two different tied sets meet in at most a segment. In (f) a pairwise intersection of dimension two requires the two atoms to share the basis in exactly one block, and sharing in both blocks would make them equal. With three pairs and two blocks, two pairs share in the same block, say (1, 2) and (1, 3) in $Q_0$; then all three atoms share the basis in $Q_0$, the pair (2, 3) shares there as well, and since no pair may share in $Q_1$ as well, all three pairwise intersections equal the diagonal states of that basis of $Q_0$ with $Q_1$ maximally mixed, which gives $M_3$.

$M_3 \oplus \mathbb{C}$ (qutrit block, outcome $a$; dimension nine). The forms of dimension at least three are: the qutrit block alone, a single set; the block set of a line $\ell$ together with $a$, dimension five; the block set of $\ell$ without $a$, dimension four; the states supported on a plane $S$ together with $a$, or with a fixed mixture of $a$ and the line orthogonal to $S$, dimension four; the states on $S$, dimension three; and the diagonal states in a basis together with $a$, dimension three. Two planes meet in a line, so the forms built on planes have pairwise intersections of dimension at most one. Block sets without a need pairwise orthogonal lines for intersections of dimension two, which makes all intersections equal to the diagonal states of one basis. Bases with pairwise intersections of dimension two must share exactly one line pairwise, and three such shared lines are mutually orthogonal and force the three bases to coincide. The block sets of

lines together with a remain. Their atoms contain the full-rank state I/4 in their fixed sets and are therefore the trace-preserving conditional expectations, the pinchings of the qutrit block. If all three lines are orthogonal, the atoms generate $M_3$. Otherwise two lines, relabelled $\ell_1$ and $\ell_2$, are not orthogonal, and the restrictions of the pinchings to the span $L$ of the orbit are commuting projections, and distinct products $x_1x_2 \neq x_1x_2x_3$ require a non-zero $Y$ in $L$ that commutes with $P_1$ and $P_2$ and satisfies $P_3YP_3 = 0$ and $(1 - P_3)Y(1 - P_3) = 0$. The commutant of $P_1$ and $P_2$ is spanned by the projector onto their span and the projector onto its orthogonal line, so $Y$ has eigenvalues $\alpha$, $\alpha$ and $\beta$, while an operator that is off-diagonal with respect to $P_3$ has a spectrum symmetric under sign change; hence $\alpha = \beta = 0$. hybrid_checks.py confirms that this sector is zero for random triples of lines in which $\ell_1$ and $\ell_2$ are not orthogonal. This step does not use the symmetry.

Table S2. Fixed-set forms of dimension at least three in the hybrids with four perfectly distinguishable states, the size of their orbit under the automorphisms, and the condition of the proof of Theorem 3(iv) that excludes them (pairwise intersections of dimension at least two that differ from each other, common intersection of dimension at least one, three forms in one orbit). $Q$, $Q_0$, $Q_1$ denote qubit blocks, $a$, $b$ classical outcomes.

| **Algebra** | **Form** | **Dimension** | **Orbit** | **Excluded because** |
|---|---|---|---|---|
| $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ | $Q \oplus a \oplus b$ | 5 | 1 | identity |
| $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ | $Q \oplus a$, $Q \oplus b$, $Q \oplus$ mixture of $a$ and $b$ | 4 | $\leq 2$ | orbit too small |
| $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ | $Q$ alone | 3 | 1 | orbit too small |
| $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ | diagonal in a basis of $Q$, $\oplus a \oplus b$ | 3 | infinite | pairwise intersections equal |
| $M_2 \oplus M_2$ | (a) one block whole, other diagonal | 5 | infinite | same block: intersections equal; mixed: reduces to (f) |
| $M_2 \oplus M_2$ | (b) one block whole, other on a line | 4 | infinite | pairwise intersections equal |
| $M_2 \oplus M_2$ | (c) one block whole, other in a fixed state | 4 | infinite | pairwise intersections equal |
| $M_2 \oplus M_2$ | (d) one block alone | 3 | 2 | orbit too small |
| $M_2 \oplus M_2$ | (e) tied $p\,\sigma \oplus (1 - p)\,V\sigma V^\dagger$ | 3 | infinite | intersections of dimension $\leq 1$ |
| $M_2 \oplus M_2$ | (f) diagonal in a basis of each block | 3 | infinite | pairs cannot share in distinct blocks |
| $M_3 \oplus \mathbb{C}$ | qutrit block alone | 8 | 1 | orbit too small |
| $M_3 \oplus \mathbb{C}$ | block set of a line $\oplus a$ | 5 | infinite | sector argument (pinchings) |
| $M_3 \oplus \mathbb{C}$ | block set of a line, without $a$ | 4 | infinite | needs orthogonal lines, then intersections equal |
| $M_3 \oplus \mathbb{C}$ | states on a plane $\oplus a$, or $\oplus$ mixture | 4 | infinite | intersections of dimension $\leq 1$ |

| | | | | |
|---|---|---|---|---|
| $M_3 \oplus \mathbb{C}$ | states on a plane | 3 | infinite | intersections are points |
| $M_3 \oplus \mathbb{C}$ | diagonal in a basis $\oplus$ $a$ | 3 | infinite | shared lines force equal bases |

The script hybrid_forms.py builds each form as the real span of its states, verifies the dimensions in Table S2, and tests random triples in each orbit together with the special alignments that maximise the intersections (shared axes, orthogonal lines, shared lines); no triple other than the pinchings of $M_3 \oplus \mathbb{C}$ and the mixed-label triples of type (a) meets the dimension conditions, and these two are excluded by the sector argument and the label argument above.

Without the symmetry. The exclusions for the three hybrids rely on the orbit condition. Without it, each hybrid carries a faithful $\mathbb{N}_0 \times B_3$. For $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ the atoms are the dephasings of the qubit block along $z$ and $x$ and the classical transfer of $b$ to $a$, and the grading partially depolarizes the qubit block with ratio $q$, moves weight $f$ from the qubit block to $a$ and weight $g$ from $a$ and from $b$ to the maximally mixed state of the qubit block. For $M_2 \oplus M_2$ the atoms are $D_Z$ in the first block, $D_Z$ in the second and $D_X$ in both, with depolarization in each block and exchange of weight between the blocks as the grading. For $M_3 \oplus \mathbb{C}$, composing each map of the $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ example with the pinching of the qutrit block onto $\mathbb{C}^2 \oplus \mathbb{C}$, which identifies $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ with a subalgebra of $M_3 \oplus \mathbb{C}$, gives idempotent, commuting channels with the same action on states in that subalgebra. hybrid_nosym.py checks idempotence, commutation, complete positivity, distinct images for $n < 6$ and an orbit that spans the state space.

## S4. Lindblad generators

The tangent vectors of Eq. (6) of the main text follow from $L_P(Q) = -2Q$ for a Pauli operator $Q$ that anticommutes with $P$ and $L_P(Q) = 0$ otherwise, applied to the components ($a$ IZ + $b$ IX + $b$ XY + $b$ ZY)/4 of $\rho_0$. $X_1$ anticommutes only with ZY among them, $Z_1$ only with XY and $Y_1Z_2$ only with IX, while $L_\tau$ multiplies every non-identity component by −1. The matrix of the four vectors in the basis (ZY, XY, IX, IZ) is triangular after reordering, with diagonal entries $-b/2$, $-b/2$, $-b/2$ and $-a/4$, so its determinant is non-zero exactly when $ab \neq 0$. The script theory_checks.py checks this closed form, the secant identities of Eq. (5) on random states, the commutation of the four superoperators, and the action of $H$ on the generators.

## S5. Hardware protocol

Parameters. Two connected qubits chosen before submission, $q = 0.5$, the 24 words $\tau^n D_A$ with $n \in \{0, 1, 2\}$ and $A \subseteq \{X_1, Z_1, Y_1Z_2\}$, all 15 non-identity Pauli targets, 2000 shots per circuit,

barriers between preparation, Pauli layer and measurement, optimization level 1 with a fixed layout, and dynamical decoupling and twirling disabled in the sampler. The 240 circuits are submitted as one job in an order fixed by the seed 20261102.

Exact mixture. Each word is a convex combination of the sixteen Pauli layers $N$, with weights $p_w(\mathrm{N})$ obtained by composing the distributions (1/2, 1/2) over {II, $P$} for each atom and $q\,\delta_{N,\mathrm{II}} + (1 - q)/16$ for each depolarizing step. The experiment measures $f(P, N)$, the expectation value of $P$ after the layer $N$ applied to the product +1 eigenstate of $P$, for the 240 pairs $(P, N)$. The eigenvalue of $P$ for the word $w$ is $\Sigma_N\, p_w(\mathrm{N})\, f(P, N)$. In the ideal case this equals $q^n$ for $P \in C(A)$ and 0 otherwise, and the script verifies that the weights reproduce these values exactly. The normalization $f_P = f(P, \mathrm{II})$ reduces multiplicative preparation and readout bias under a factorized error model, so the criteria below are sensitive to errors that depend on the inserted layer. The 360 reported normalized eigenvalues are derived from the same 240 layer-circuit observations; they are not 360 independent circuit executions.

Acceptance criteria, fixed before data are taken. (a) Every normalized eigenvalue is within 0.08 of its prediction. (b) The eight zero-grade words have distinct kept sets, and the union law holds for all 64 pairs. (c) For every kept target, the ratio of normalized eigenvalues at $n + 1$ and $n$ equals $q$ within 0.08. (d) The 24 images of $\rho_0$ computed from the measured eigenvalues are pairwise separated by more than three combined standard errors.

Table S3. Simulation of the protocol (hw_protocol_p4.py dryrun SPAM P_GATE) with binomial shot noise, a multiplicative preparation and readout fidelity SPAM per non-identity factor of the target, and depolarization P_GATE of every non-identity gate of the inserted layer.

| **SPAM** | **P_GATE** | **(a) max deviation** | **(c) max ratio deviation** | **(d) min separation** | **Result** |
|---|---|---|---|---|---|
| 1.00 | 0 | 0.000 | 0.000 | $1.7 \times 10^4\ \sigma$ | all pass |
| 0.97 | $1 \times 10^{-3}$ | 0.013 | 0.003 | 63.2 $\sigma$ | all pass |
| 0.90 | $5 \times 10^{-3}$ | 0.033 | 0.006 | 32.4 $\sigma$ | all pass |
| 0.80 | $2 \times 10^{-2}$ | 0.048 | 0.009 | 19.7 $\sigma$ | all pass |
| 0.90 | $5 \times 10^{-2}$ | 0.083 | 0.015 | 30.5 $\sigma$ | (a) fails |
| 0.90 | $1 \times 10^{-1}$ | 0.136 | 0.030 | 28.1 $\sigma$ | (a) fails |

The 240 circuits were checked against an exact state-vector simulation, and every ideal expectation value is reproduced. The full submission path was also run on a noisy local model of a 156-qubit superconducting processor, where all criteria are met. A failure of the criteria on

hardware would indicate errors of the Pauli layers or of state preparation on the chosen qubits and would not affect Theorems 1 and 2.

Hardware run. The job dbkatgl6p51s73cm9r2g ran on ibm_kingston on 2 November 2026 (submitted 15:39:47 UTC, finished 15:45:30 UTC), on physical qubits 148 and 149, with qiskit 2.6.1 and qiskit-ibm-runtime 0.51.0 and 128 s of processor time. The qubit pair was chosen before submission as the connected pair with the smallest sum of readout and single-qubit gate errors. No data were excluded.

Table S4. Results of the protocol on ibm_kingston (qubits 148, 149), 240 circuits with 2000 shots each.

| **Quantity** | **Value** | **Criterion** | **Result** |
|---|---|---|---|
| Identity-layer factors $f_P$ | 0.985 to 1.000 | none | |
| Signs of $f(P, N)$ matching commutation | 240 of 240 | none | |
| min over layers of $\lvert f(P, N)\rvert/f_P$ | 0.978 | none | |
| (a) max deviation of 360 normalized eigenvalues | 0.009 (mean 0.002) | $\leq 0.08$ | pass |
| Removed components, max magnitude | 0.009 | none | |
| Kept components at $n = 0$, min value | 0.994 | none | |
| (b) distinct kept sets and union law | 8 distinct, 64 of 64 pairs | all | pass |
| (c) max deviation of grading ratio from $q$ | 0.004 | $\leq 0.08$ | pass |
| (d) min separation of the 24 images of $\rho_0$ | 146 $\sigma$ | $> 3\ \sigma$ | pass |

Symmetry test. A second job (dbkb7bhffo6c73bb62mg) ran on the same qubits on 2 November 2026 (submitted 16:00:46 UTC, finished 16:07:04 UTC) with 167 s of processor time and 315 circuits of 2000 shots, in an order fixed by the seed 20261103. Part A prepares each of the four eigenvectors of $\rho_0$, or the control state $|0\rangle|+\rangle$, by a state-preparation circuit, applies nothing, $U_C$ or $U_T$, and measures in the nine product Pauli bases; the Pauli vector of $\rho_0$ is the eigenvalue-weighted mixture of the four reconstructions. Part *B* runs, for $g \in \{U_C, U_T\}$ and

each atom $x_i$ with Pauli operator $P_i$, the circuits $g^\dagger$, $N$, $g$ with $N \in \{\mathrm{II}, P_i\}$ on the +1 eigenstate of each of the 15 targets, and forms $\lambda_P = (E_{\mathrm{II}} + E_{\mathrm{Pi}})/(2E_{\mathrm{II}})$. The prediction is $\lambda_P = 1$ if $P$ commutes with $g\ P_i\ g^\dagger$ and 0 otherwise. The Clifford circuits are synthesized from the tableaux of Eq. (3) and contain at most two CZ gates after transpilation. The replication on ibm_fez (job dbkbevonceqs73ali8gg, submitted 16:17 UTC, executed 17:17–17:22 UTC, 166 s of processor time) used the same circuits, seed and criteria on qubits 109 and 118, chosen before submission as the connected pair with the smallest readout and gate errors. Acceptance criteria, fixed before submission: (e) the components of $\rho_0$ change by at most 0.08 under $U_{\mathrm{C}}$ and $U_{\mathrm{T}}$; (f) the control state lands within 0.15 of its predicted image and moves by at least 0.5; (g) every $\lambda_P$ is within 0.10 of its prediction, and the kept sets reproduce the full $S_3$.

Table S5. Results of the symmetry test, 315 circuits with 2000 shots each, on ibm_kingston (qubits 148, 149) and on ibm_fez (qubits 109, 118).

| **Quantity** | **ibm_kingston** | **ibm_fez** | **Criterion** | **Result** |
|---|---|---|---|---|
| $\rho_0$ as prepared, max deviation from Eq. (4) | 0.025 | 0.042 | none | |
| (e) max change of $\rho_0$ under $U_{\mathrm{C}}$, $U_{\mathrm{T}}$ | 0.023, 0.033 | 0.033, 0.045 | ≤ 0.08 | pass |
| (f) control: max deviation from predicted image under $U_{\mathrm{C}}$, $U_{\mathrm{T}}$ | 0.089, 0.036 | 0.073, 0.043 | ≤ 0.15 | pass |
| (f) control: largest change under $U_{\mathrm{C}}$, $U_{\mathrm{T}}$ | 1.01, 0.98 | 1.03, 1.01 | ≥ 0.5 | pass |
| (g) max deviation of the 90 conjugated-channel eigenvalues | 0.021 | 0.039 | ≤ 0.10 | pass |
| (g) induced permutations of $(x_1, x_2, x_3)$ | $U_{\mathrm{C}}$: cycle; $U_{\mathrm{T}}$: $x_1 \leftrightarrow x_2$ | same | full $S_3$ | pass |

Composition test. comp_core.py lists, for each of the six elements, every sequence of Pauli layers with its weight: II or $P_i$ with weight 1/2 for each atom, and II with weight $q + (1 - q)/16$ or any other of the sixteen layers with weight $(1 - q)/16$ for each depolarizing step. Every

sequence is run for each eigenvector of $\rho_0$ and for the settings ZZ, ZX, XY and ZY, with shots proportional to its weight (1000 per element, eigenvector and setting, and at least 8 per circuit), 2544 circuits in all. The measured component is the weighted sum over sequences and eigenvectors. Criteria fixed before submission: (h) every one of the 24 components within 0.06 of its prediction; (i) every pair of the six images separated by more than three combined standard errors in some component.

Table S6. Results of the composition test, 2544 circuits and about $10^5$ shots in total.

| **Quantity** | **ibm_kingston (148, 149)** | **ibm_fez (84, 85)** | **Criterion** | **Result** |
|---|---|---|---|---|
| Job | dbkp0adgg49c73djcvm0 | dbkp460nceqs73amco70 | | |
| (h) max deviation of 24 components | 0.024 | 0.033 | ≤ 0.06 | pass, pass |
| (i) min pairwise separation | 4.5 $\sigma$ | 2.7 $\sigma$ | > 3 $\sigma$ | pass, fail |
| Weakest pair | $\tau$ versus $\tau x_1x_2x_3$ | $x_1x_2x_3$ versus $\tau x_1x_2x_3$ | | |
| IZ of $x_1x_2x_3$ and $\tau x_1x_2x_3$ (predicted 0.2, 0.1) | 0.207, 0.100 | 0.167, 0.114 | | |

Follow-up on ibm_fez (job dbkpb95gg49c73djddtg, qubits 84 and 85, executed 08:07–08:11 UTC on 3 November 2026). After the failure of criterion (i) on ibm_fez, a separate test was defined before submission (comp6_core.py): only the elements $x_1x_2x_3$ and $\tau x_1x_2x_3$, only the setting ZZ, 8000 shots per element and eigenvector, 544 circuits. Criteria: both IZ values within 0.06 of 0.2 and 0.1, and a separation of more than three combined standard errors. Measured: 0.196 and 0.117, maximal deviation 0.017, separation 10.7 standard errors; both criteria are met.

## S6. Reproducibility

theory_checks.py runs the checks of Sections S1 to S4 and prints PASS or FAIL for each. product_observer.py constructs the product symmetry group of S3. hw_protocol_p4.py builds the 240 circuits, simulates the protocol with optional noise (mode dryrun), submits it to an IBM Quantum backend (mode submit, which needs qiskit and qiskit-ibm-runtime), retrieves the results (mode retrieve) and applies the acceptance criteria (mode analyze). P4_IBM_run.ipynb runs the same protocol from a notebook and asks for the access key at run

time; P4_IBM_observer_test.ipynb and obs_core.py run and analyze the symmetry test. The raw counts and specifications of all six hardware runs are in the data folders of the repository, one folder per run. All files are in the repository cited in the main text [S3]. The checks need NumPy only.

**References for the Supplementary Material**

[S1] A. Fujiwara, P. Algoet, One-to-one parametrization of quantum channels, Phys. Rev. A 59 (1999) 3290–3294. doi:10.1103/PhysRevA.59.3290.

[S2] M. B. Ruskai, S. Szarek, E. Werner, An analysis of completely-positive trace-preserving maps on $M_2$, Linear Algebra Appl. 347 (2002) 159–187. doi:10.1016/S0024-3795(01)00547-X.

[S3] A. M. Tishin, Code and data repository, https://github.com/alexandermtishin-art/Code-and-IBM-Quantum-Data (2026).